

Exam Fall 2024

The Duration of the exam is 120 minutes.

The exam will be delivered in digital form as a set of 50 true-false questions:

- You will download a PDF document with the questions
- You will also download an **empty spreadsheet, called also the “answer sheet”**, where the question numbers are also reported, this is the answer document you need to deliver
- You will mark on the **spreadsheet** your answer next to the corresponding question number
- For each question mark in the answer sheet your answer, only write True or False, or you can leave the cell blank if unsure
- No note or comment can be added to it in any part, violating this will risk invalidating the entire question or group of questions!
- You **will deliver the spreadsheet** with your answer in in XLSX, CSV, or ODT formats

During the exam you can use pen and paper to note down anything you may need or to help yourself calculate your answer.

During the exam you can ONLY use the standard calculator ON YOUR LAPTOP.

Use of phone, even for calculator only, is not allowed under any circumstance.

During the exam no aid, notes, book, or any other support material can be used.

It is NOT allowed to seek help from or consult others (human beings, artificial intelligences, websites, DBMSs, other software, etc.) during the exam.

Marking and Grades

- Each **correct** answer will assign **+2 points**
- Each **wrong** answer will assign **0 points**
- Leaving the answer **empty/blank will assign 0 points**
- The total maximum amount of points for the entire exam is 100

Group A: Entity Relationship (ER) Modelling [20 points total]

Question A.01 [2 Points]: Relational database management systems store all data using graph structures.

(T) True (F) False

Question A.02 [2 Points]: A relation in an ER diagram may have its own attributes.

(T) True (F) False

Question A.03 [2 Points]: When converting an ER diagram into a set of tables, every relationship in the ER diagram may not lead to its own table.

(T) True (F) False

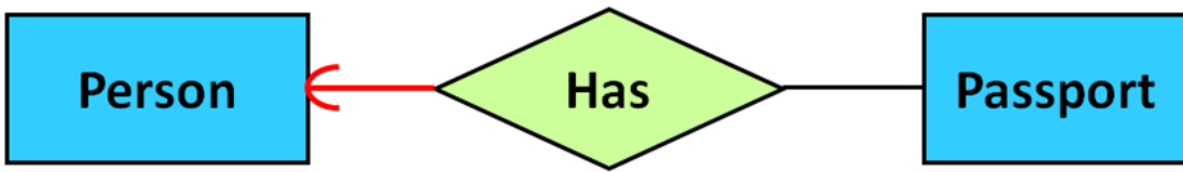
Question A.04 [2 Points]: A subclass entity set in an ER diagram may not inherit the same primary key from its parent entity set.

(T) True (F) False

Question A.05 [2 Points]: When converting an ER diagram into a set of tables, a weak relationship in the ER diagram never leads to its own table.

(T) True (F) False

Consider the following ER diagram (Questions A.06-A.10). For one-to-many relations, the 'one'-side has an arrow and the 'many'-side has no arrow. Referential integrity constraint is denoted by a rounded arrow instead of a pointed arrow. Pointed arrow is denoted as . Rounded arrow is denoted as . That is, in the following example, every "Passport" must be involved in the "Has" relationship.



Question A.06 [2 Points]: Based on the above ER diagram, each person can have at most one passport.

(T) True (F) False

Question A.07 [2 Points]: Based on the above ER diagram, each passport can be possessed by at most one person.

(T) True (F) False

Question A.08 [2 Points]: Based on the above ER diagram, there can be a passport that is not possessed by any person.

(T) True (F) False

Question A.09 [2 Points]: Based on the above ER diagram, there can be a person not having any passport.

(T) True (F) False

Question A.10 [2 Points]: When converting the above ER diagram into a set of tables, the primary key attribute of the 'person' entity set will be included as a non-key attribute in the table corresponding to the 'passport' entity set.

(T) True (F) False

Group B: Normalization [20 points total]

Question B.01 [2 Points]: A table with three attributes and no non-trivial functional dependency (FD) is in BCNF.

(T) True (F) False

Question B.02 [2 Points]: Given a set of functional dependencies, the minimal basis may not be unique.

(T) True (F) False

Question B.03 [2 Points]: If a table satisfies the 3rd normal form, it must also satisfy the BCNF.

(T) True (F) False

Question B.04 [2 Points]: A table with two attributes always satisfies the 2nd normal form.

(T) True (F) False

Question B.05 [2 Points]: A BCNF decomposition always preserves all functional dependencies.

(T) True (F) False

Question B.06 [2 Points]: $R(A, B, C, D, E)$ is a relation schema and $F = \{A \rightarrow BC, AB \rightarrow D, B \rightarrow CE\}$ are functional dependencies on R . Then, $\{A, B, C, E\}$ is the closure of the attribute set $\{B\}$.

(T) True (F) False

Question B.07 [2 Points]: $R(A, B, C, D, E)$ is a relation schema and $F = \{E \rightarrow D, D \rightarrow A, ADE \rightarrow B, C \rightarrow AB\}$ are functional dependencies on R . CE is the only candidate key of R .

(T) True (F) False

Question B.08 [2 Points]: $R(A, B, C, D, E)$ is a relation schema and $F = \{E \rightarrow D, D \rightarrow A, ADE \rightarrow B, C \rightarrow AB\}$ are functional dependencies on R . Then, R is in BCNF.

(T) True (F) False

Question B.09 [2 Points]: $R(A, B, C, D, E)$ is a relation schema and $F = \{E \rightarrow D, D \rightarrow A, ADE \rightarrow B, C \rightarrow AB\}$ are functional dependencies on R . Then, R is in 3NF.

(T) True (F) False

Question B.10 [2 Points]: $R(A, B, C, D, E)$ is a relation schema and $F = \{E \rightarrow D, D \rightarrow A, ADE \rightarrow B, C \rightarrow AB\}$ are functional dependencies (FDs) on R . Consider the decomposition of R into $R_1(D, A)$, $R_2(C, B)$, $R_3(E, D)$, $R_4(C, E)$. This decomposition preserves all FDs in F .

(T) True (F) False

Group C: Relational Algebra [10 points total]

σ (selection), Π (projection), $\cup$ (union), $\cap$ (intersection), $-$ (difference), γ (grouping and aggregation), δ (duplicate elimination), $\div$ (division), $:=$ (assignment), ρ (rename), $\bowtie$ (natural join), $\bowtie_{\text{condition}}$ (theta join), $\times$ (cartesian product)

Question C.01 [2 Points]: Intersection operation in relational algebra with sets semantics eliminates duplicate tuples.

(T) True (F) False

For Questions C.02-C.03, consider a database schema as follows. Primary Keys are in **bold**.

TRAINS (**train_num**, source_city, destination_city)

PASSENGERS (**passenger_id**, passenger_name, address)

RESERVATIONS (**passenger_id**, **train_num**, **date**, seat_number)

Question C.02 [2 Points]: The 'Result' relation below returns the names of the passengers who have never reserved a train going from Aalborg to Aarhus.

$R1 := \sigma_{\text{source_city}='Aalborg' \text{ AND } \text{destination_city}='Aarhus'} \text{TRAINS}$

$R2 := \Pi_{\text{passenger_id}}(R1 \bowtie \text{RESERVATIONS})$

$\text{Result} := (\Pi_{\text{passenger_id}} \text{PASSENGERS}) - R2$

(T) True (F) False

Question C.03 [2 Points]: The 'Result' relation below returns the train(s) that has been reserved by all passengers living in Copenhagen (regardless of the date).

$R1 := \Pi_{\text{passenger_id}} (\sigma_{\text{address}='Copenhagen'} \text{PASSENGERS})$

$R2 := \Pi_{\text{passenger_id}, \text{train_num}} \text{RESERVATIONS}$

$\text{Result} := R2 \div R1$

(T) True (F) False

For Questions C.04-C.05, consider a database schema as follows. Primary Keys are in **bold**.

Student (**snum**, sname, major, level, age)

Class (**cname**, time, room, fid)

Enrolled (**snum**, **cname**)

Faculty (**fid**, fname, school)

Question C.04 [2 Points]: The 'Result' relation below returns the name(s) of the class(s) that has never been enrolled by students whose major is 'CSE'.

$R1 := \Pi_{cname} \text{ Class}$

$R2 := \Pi_{cname} (\text{Enrolled} \bowtie (\sigma_{major='CSE'} \text{ Student}))$

$\text{Result} := R1 - R2$

(T) True (F) False

Question C.05 [2 Points]: The 'Result' relation below returns the faculty member(s) who has taught the maximum number of courses.

$R1 := \gamma_{fid, COUNT(cname) \rightarrow CourseNum} \text{ Class}$

$R2 := \gamma_{MAX(CourseNum) \rightarrow MaxNum} \text{ Class}$

$R3 := \Pi_{fid, CourseNum} (R1 \bowtie_{R1.CourseNum < R2.MaxNum} R2)$

$R4 := \gamma_{MAX(CourseNum) \rightarrow MaxNum} R3$

$\text{Result} := \Pi_{fid} (R3 \bowtie_{R3.CourseNum = R4.MaxNum} R4)$

(T) True (F) False

Group D. Basic SQL [10 points]

Consider the following tables for questions in Groups D and E.

Genre

genreid	genrename
1	Sci-Fi
2	Crime
3	Animation
4	Romance
5	Action
6	Western

Movie

movieid	title	releaseyear	director	rating
1	Inception	2010	Christopher Nolan	3
2	Interstellar	2014	Christopher Nolan	4
3	The Godfather	1972	Francis Ford Coppola	3
4	Expendables	NULL	NULL	1
5	Spirited Away	2001	Hayao Miyazaki	2
6	Titanic	1997	NULL	3

Viewer

viewerid	firstname	lastname	email
1	Alice	Johnson	alice@example.com
2	Bob	Smith	NULL
3	Carol	Lee	carol@example.com
4	David	Brown	NULL
5	Juan	NULL	juan@example.com
6	Eve	Wilson	eve@example.com

MovieGenre

movieid	genreid
1	1
2	1
3	2
5	3
6	4
1	4
4	5

Rating

movieid	viewerid	rating	review
1	2	5	A fantastic blend of science and storytelling.
2	1	4	Great visuals but a bit long.
3	3	5	The best mafia movie ever made.
5	4	3	Not my style, but well-made.
6	6	NULL	NULL
2	5	2	Confusing at times.
4	3	NULL	Skipped this movie.
1	6	5	Inception is mind-blowing!
5	1	4	Magical animation with a deep story.
3	4	5	An unforgettable classic.
6	2	4	Beautifully shot, but the story felt flat.
4	5	NULL	Haven't seen it yet.
2	3	3	Good, but not my favorite.
1	4	5	Watched it multiple times!
5	2	NULL	No review for this one.

Question D.01 [2 points]: The following query returns two tuples.

```
SELECT * FROM Movie WHERE director = NULL;
```


(T)True (F) False

Question D.02 [2 points]: The following query returns no tuple for movieid = 4.

```
SELECT movieid, AVG(rating) AS rt
FROM Rating
GROUP BY movieid HAVING AVG(rating) > 2;
```

(T)True (F) False

Question D.03 [2 points]: The following query returns one tuple.

```
SELECT movieid, rating FROM Movie NATURAL JOIN Rating;
```

(T)True (F) False

Question D.04 [2 points]: The following query returns tuple (6, 2, 4) among others.

```
SELECT r1.movieid, r1.viewerid, r2.viewerid
FROM Rating as r1
JOIN Rating as r2
ON r1.movieid=r2.movieid
AND r1.rating=r2.rating;
```

(T)True (F) False

Question D.05 [2 points]: The following query returns 5, despite being 6 tuples in the table Movie.

```
SELECT COUNT(*) FROM Movie
WHERE director = director OR releaseyear=releaseyear;
```

(T)True (F) False

Group E. Advance SQL [10 points]

Question E.01 [2 points]: The following query returns NO tuple with the genre name "Western".

```
SELECT g.genrename, m.title
```

```
FROM MovieGenre as mg JOIN Movie as m ON mg.movieid = m.movieid
RIGHT OUTER JOIN Genre AS g ON g.genreid = mg.genreid;
```

(T)True (F) False

Question E.02 [2 points]: The following query returns two tuples for viewerid = 1.

```
SELECT r.viewerid, r.rating, m.title
FROM Rating as r INNER JOIN Movie as m ON r.movieid = m.movieid
WHERE r.rating <= ALL(SELECT r1.rating
                     FROM Rating as r1
                     WHERE r1.viewerid = r.viewerid
                     AND r1.rating IS NOT NULL);
```

(T)True (F) False

Question E.03 [2 points]: The following query is not a valid query as “v.email IS NULL” cannot be placed in the HAVING clause and should be placed in the WHERE clause.

```
SELECT v.email, AVG(r.rating)
FROM Viewer as v FULL OUTER JOIN Rating as r ON v.viewerid = r.viewerid
GROUP BY v.email HAVING v.email IS NULL;
```

(T)True (F) False

Question E.04 [2 points]: The following query returns six tuples because there are six movies and six reviewers.

```
WITH AvgViewerRating AS(
    SELECT viewerid, AVG(rating) AS vrating
    FROM Rating GROUP BY viewerid
), AvgMovieRating AS (
    SELECT movieid, AVG(rating) AS mrating
    FROM Rating GROUP BY movieid
)
SELECT vr.viewerid, mr.movieid, vr.vrating/mr.mrating AS rate
FROM AvgViewerRating AS vr JOIN Rating r ON vr.viewerid = r.viewerid
    JOIN AvgMovieRating AS mr ON r.movieid = mr.movieid
    WHERE vr.viewerid=mr.movieid;
```

(T)True (F) False

Question E.05 [2 points]: The column “count” resulting from executing the following query only contains zeros.

```

SELECT r.movieid, COUNT(m.title) AS count
FROM Rating as r LEFT JOIN Movie as m
      ON r.movieid=m.movieid AND m.rating IS NULL
GROUP BY r.movieid;

```

(T)True (F) False

Group F. DDL [10 points]

Consider the following table definition for questions in group F.

<pre> CREATE TABLE Movie (movieid PRIMARY KEY, title VARCHAR(255), releaseyear INT, director VARCHAR(255), rating INT CHECK (rating BETWEEN 1 AND 5)); CREATE TABLE Viewer (viewerid PRIMARY KEY, firstname VARCHAR(255), lastname VARCHAR(255), email VARCHAR(255)); CREATE TABLE Genre (genreid PRIMARY KEY, genrename VARCHAR(255)); </pre>	<pre> CREATE TABLE MovieGenre (movieid INT REFERENCES Movie(movieid) ON DELETE CASCADE, genreid INT REFERENCES Genre(genreid), PRIMARY KEY (movieid, genreid)); CREATE TABLE Rating (movieid INT REFERENCES Movie(movieid) ON DELETE CASCADE, viewerid INT REFERENCES Viewer(viewerid) ON DELETE RESTRICT, rating INT CHECK (rating BETWEEN 1 AND 5), review TEXT, PRIMARY KEY (MovieID, ViewerID)); </pre>
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Question F.01 [2 points]: It is not possible to delete a “Movie” if there is at least one tuple in either “Rating” or “MovieGenre” that refers to it.

(T)True (F) False

Question F.02 [2 points]: the “Rating” table can contain a tuple with a NULL value for the “movieid”.

(T)True (F) False

Question F.03 [2 points]: the “Movie” table can contain a tuple with a NULL value for the “title”.

(T)True (F) False

Question F.04 [2 points]: if the “Genre” table can contain a tuple with genreid=1, then the table “MovieGenre” necessarily exist at least a tuple with genreid=1.

(T)True (F) False

Question F.05 [2 points]: the following create statement will fail generate an error.

```
CREATE TABLE RatingDescription (  
    rating int REFERENCES movie(rating),  
    description text,  
    PRIMARY KEY(rating)  
);
```

(T)True (F) False

Group G: Storage representation [10 points]

Question G.01 [2 points]: Database storage is organized in pages, usually between 512b. and 16kb.

(T)True (F) False

Question G.02 [2 points]: k-way External Merge Sort only requires reading and writing each page only once.

(T)True (F) False

Question G.03 [2 points]: Since in practice B+tree nodes are pages, databases tend to use B+Tree of orders usually lower than 8, i.e., a node have at most 8 children.

(T)True (F) False

Question G.04 [2 points]: B+trees cannot be used for improving performance on exact matching queries.

(T)True (F) False

Question G.05 [2 points]: k-way External Merge Sort allows to sort relations that are larger than (RAM) memory.

(T)True (F) False

Group H: Query processing & Transactions [10 points]

Question H.01 [2 points]: Given a table T(a, b, c) with a B+tree index on (b, c), such index is useful to solve queries such as “SELECT * FROM T WHERE b<10”, “SELECT * FROM T

WHERE $b > 10$ AND $b < 25$ ” or “SELECT * FROM T WHERE $b \geq 5$ AND $c < 10$ ”. Yet, such index is NOT useful for the query “SELECT * FROM T WHERE $c < 5$ ”

(T)True (F) False

Question H.02 [2 points]: The block nested join requires reading each page of the inner table as many times as there are pages in the outer table.

(T)True (F) False

Question H.03 [2 points]: The query optimizer ALWAYS returns the BEST possible plan.

(T)True (F) False

Question H.04 [2 points]: In a Two-phase Locking, locks are acquired and release in any order.

(T)True (F) False

Question H.05 [2 points]: Both schedule S1 and S2 are serializable, but they are not equivalent.

S1	S2
R(A)	
	W(A)
	R(B)
R(B)	
	W(B)
W(B)	

S1	S2
	W(A)
R(A)	
	R(B)
	W(B)
R(B)	
W(B)	

(T)True (F) False